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Meanwhile, the growing demand for Information Technology (IT) which can help the management and organisation of e-learning led to the development of Learning Management Systems (LMS). The LMS seek to automate the administration of the courses, to record users, to record courses, to record information about the learning process and to provide reports to the course administration (Coutinho, 2009). E-learning implies important changes in the culture and structure of the institutions that decide to adopt it (Moore and Kearsley, 2007). It also assists in the production of new knowledge backed by IT, and an integrated view of it can enable the creation and management of internal and external processes as parts of a great organisational system (Vieira *et al.*, 2005).

E-learning gives students accessibility to data; wherever one looks, data of e-learning is there. With the use of mobile devices as well as many other IT gadgets, people around the globe anytime, anywhere have access to data. This improves interactions between students and their instructors for their classes. Also, different groups of students can study independently, without interfering with one another. The number of communication tools, offered by a Learning Management System, such as a chat, a forum, social communities, blogs, wikis etc. allows a teacher to engage students and give a voice to less confident ones.

1.2 Research Objectives

The objectives of this research work are

- a. To develop a web-based for teaching and learning.
- b. To deliver course content in an organised manner in order to enhance the competence and performance of students.

CHAPTER ONE

INTRODUCTION

The use of the e-learning systems is gaining more popularity in the modern higher education experience in Nigeria and the rest of the world. Whether distance or traditional student, residential or non-residential campus, undergraduate or post-graduate, the Learning Management System is being rapidly accepted in higher education. In the past few years, online course management systems have replaced other alternative means to deliver class contents such as live satellite or closed-circuit television. The introduction of learning management systems, along with increased computer usage in homes and in businesses has brought an increasing number of students and teachers to the online learning environment (Falvo and Johnson, 2007).

1.1 Background of the Study

E-learning is a result of significant changes in the definition of education concept, as well as changes in the understanding of how it should be organised and managed (Peters, 2003). With the advance of e-learning, educational institutions managers started to deal with different activities, requiring the development of new procedures and finding alternatives to address emerging challenges that go beyond educational issues. According to Moore and Kearsley (2007), e-learning is a planned learning process that occurs in general, in a different medium other than a regular school, and as a result, it requires special techniques of course design, special forms of instruction, special methods of communication through electronic and other technologies, as well as essential organisational and administrative arrangements.

CHAPTER TWO

LITERATURE REVIEW

2.1 Information Technology (IT)

IT can be an important component of innovation, not only by its direct application, but also because it is a vector for other innovations that facilitates, enhances, and, among others, highlights e-learning (Laurindo, 2016). In the scenario of globalisation, in which virtual organisations and e-businesses develop, there are great expectations regarding the potential of Information Technology, which increases the importance of analysis of its role. However, when using IT in the educational process, it is essential to identify the conceptions that underlie its development, having an adequate view of its possibilities and potentials, because depending of its use, it will be explicit the understanding that we have of the educational process in a space that includes the technology itself (Schlemmer, *etal*, 2007).

The development of IT has generated interactive media allowing learning and collective construction of knowledge through networks, with interchangeability of the roles of source and receiver. But the inclusion of IT in e-learning projects only began in the 1990s (Souza, 2005). In this context, the interest in thinking the interactivity offered by IT grows and its impact on education and organisational culture (Sartori and Garcia, 2009).

According to Oliveira (2012), the potential that IT offers may make e-learning closer to the classroom mode in relation to personal interaction and preserve the distance between teachers and students, in order to improve the process of mediated communication, systematic guidance and constant monitoring, focused on the formation of skills and attitudes that allow

CHAPTER THREE

METHODOLOGY

3.1 Development Approach

The agile software development methodology was employed in the development process. This is because it allows for visible results quickly and for quick iteration and advancement of the system.

3.1.1 The Agile Software Development Approach

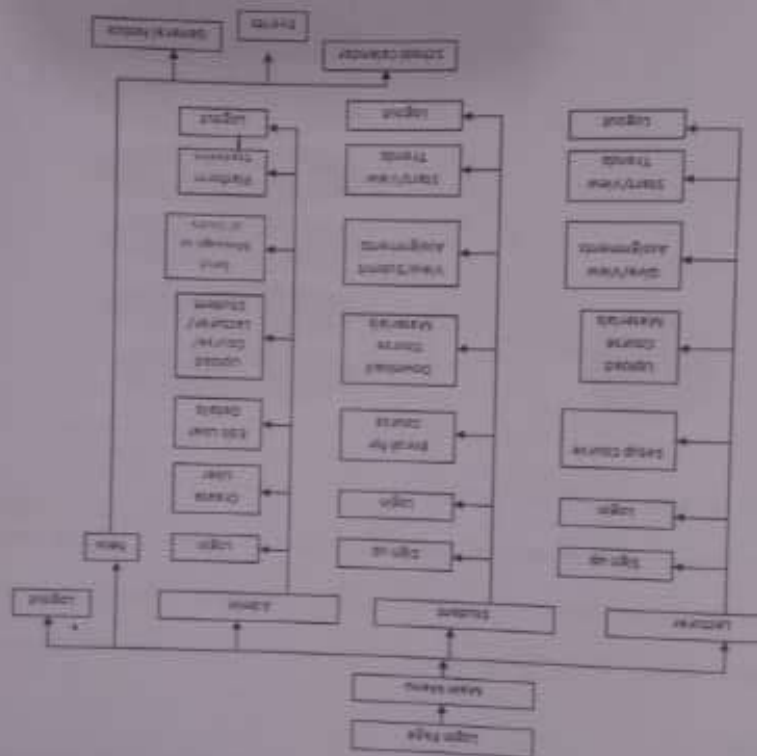
Agile is a methodology that anticipates the need for flexibility and applies a level of pragmatism into the delivery of the finished product. This methodology is rooted in adaptive planning, early delivery and continuous improvement, all with an eye toward being able to respond to change quickly and easily.

Whereas Waterfall includes several testing and bug fixing cycles before releasing a product, this methodology is far more collaborative and iterative. One of the biggest differences is that Waterfall calls for heavy documentation early on. This documentation makes it harder to change features as the process goes on. Agile significantly increases productivity and reduces time to benefits relative to the classic Waterfall processes. Agile processes enable organisations to adjust smoothly to rapidly-changing requirements, and produce a product that meets evolving goals. Agile Scrum process benefits an organisation/institution by helping to increase the quality of the deliverables, cope better with change, provide better estimates while spending less time creating them as well as to be more in control of the project schedule and state.



- c. To provide feature tools like chat rooms, communication with the students through messaging and notifications which will enable interaction between the lecturers and students as well as encourage students' participation.

Figure 4.1: Top-Down Design of the System



4.6 Use Case Diagrams

The following diagrams demonstrate the different ways that each user may interact with the system. They describe the set of actions (use cases) that a user can perform within the system.

A. Lecturer

The lecturer typically has access to sign up, login as a registered user, set up course(s), upload course materials, give assignment as well as view assignments submitted by students and start a message thread in terms of chat rooms or forums. Figure 4.2 below shows the actions that can be performed by a lecturer.

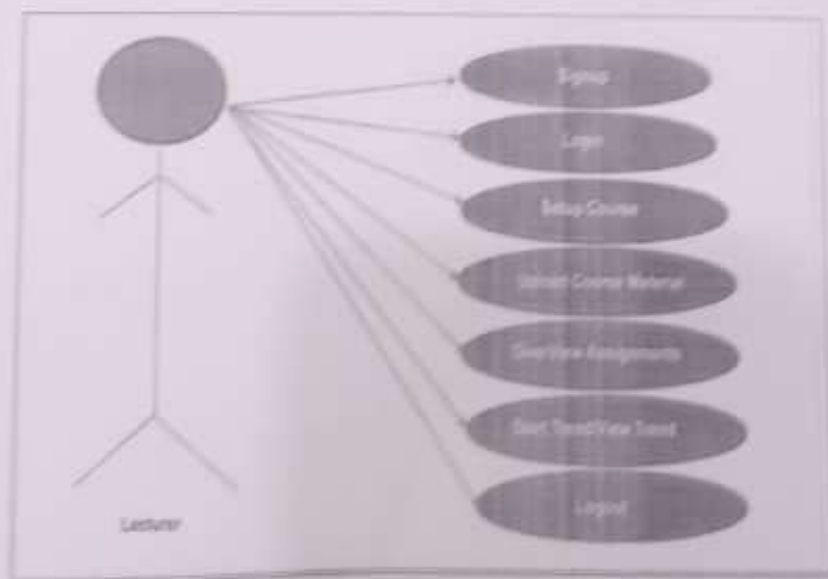


Figure 4.2: Lecturer Use Case diagram

CHAPTER FOUR

DESIGN AND IMPLEMENTATION

4.1 Features of the System

Features of the system are: login page (student/instructor/admin), register student page, register instructor page, update instructor/student page, view instructor page and view student page; also security, delivery, interaction, reporting and recordkeeping are put into place. The system requires the user to login to have access to the website to do the following tasks: view and download course materials, upload files, projects and other materials for people to see and also interact through live chats and discussion forums. Also, the LMS will allow instructors to upload videos that will be viewed and downloaded. All users are allowed to login simultaneously on multiple systems.

4.2 Development Approach

This research adopted the Object-Oriented approach. The object-oriented approach decomposes the system down into objects, then creates an object model which groups the objects into classes, and describes each class in terms of its attributes (or data), methods (or functions) and relationships to other classes.

4.3 Web-Based Technologies Employed

The web-based technologies employed for the development of the teaching and learning system include:

B. Lecturer's Desktop

The lecturer's desktop provide links to the major operations that can be performed by the lecturer. These include creation of courses sites, removal of enrolled student (if applicable), course list (administration/operation on courses can be done by clicking on them), messages, events, announcements account management and statistics. See figure 4.11 below:



Figure 4.11: Lecturer's Desktop

C. Course Site Creation (Lecturer)

On this page, a lecturer can create a course site by entering all the required course parameters. The lecturer can also implement course settings like course access (which specifies whether the course can be assessed without login or only registered students or a particular user list), registration validation (which requires students who intends to register

4.7 Program Flow

The program flow relates to all groups of users. Every login attempt is verified by the system. An unregistered user is directed to register before being able to login. Upon successful registration the user is permitted to login and have access to the user menu with respect to the user group. A registered user is required to provide the username and password. Successful verification gives access to the user into the system. Unsuccessful verification returns the user back to the login page. Figure 4.5 below describes the program flow of the system.

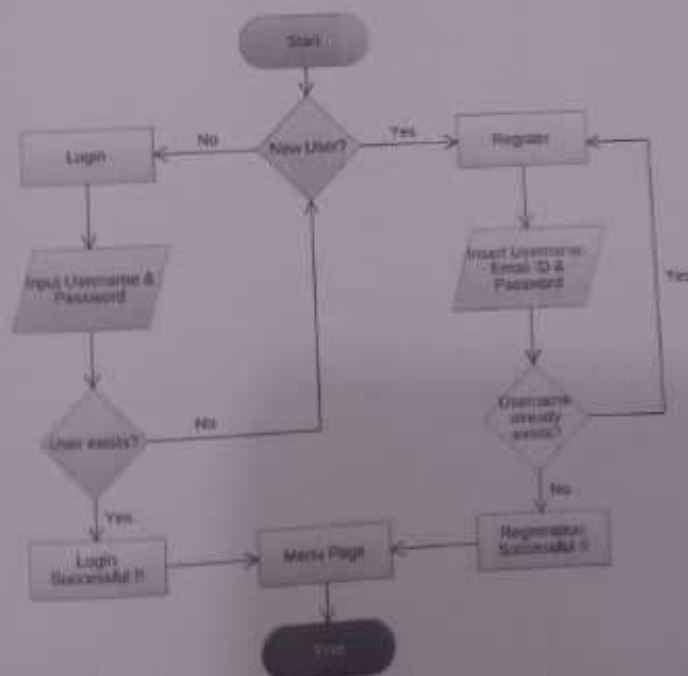


Figure 4.5: Program Flow Chart of the System

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

This project is on design and development of a teaching and learning system for higher learning. The system was developed as a web application, which can be accessed with any browser on any operating system and was hosted on localhost.

The designed learning system was first implemented on three students. The chosen students had different perception of the e-learning system. After the test, these three students were asked to complete questionnaires. The questionnaires were then used to measure students' responses in e-learning process. Based on the analysis of completed questionnaires, the mean score of students' responses was 3.5 or 70%. It showed that the students had positive responses to the software. Because of these positive responses, the system was then implemented on a small group. In the small group implementation, six students were chosen who also had different perception of the e-learning system. After the test, these students were asked to complete questionnaires. The questionnaires were again used to measure students' responses. The analysis of completed questionnaires brought about a mean score of students responses of 4.01 or 78%. This also showed that the students had positive responses to the software. These positive responses showed that the developed teaching and learning system was effective and valid.

This system will aid the delivery of course contents in an organised manner which will enhance the competence and performance of students. It will also help automate students' registration process while providing feature tools like chat rooms, communication with the students through messaging and notifications to enable interaction between the lecturers and students as well as encourage students' participation.